

AGILE PROJECT DOCUMENTATION

AI Career Companion Agent for Internship Matching and Interview Preparation

Prepared By

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Project Duration

15 July – 9 September

Project Repository

github.com/MallarpuAishwarya/AI-Career-Companion-Agent-for-Internship-Matching-and-Interview-Preparation

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1. Project Overview

1.1 Project Name

AI Career Companion Agent for Internship Matching and Interview Preparation

1.2 What is the Project?

The project is an AI-powered career assistant that helps students and early-career job seekers discover internships that match their skills and profile, and prepares them for interviews through AI-generated, role-specific questions and feedback. It brings internship discovery, resume/skill matching, and interview practice into a single conversational workflow.

1.3 Problem Statement

Students and freshers spend a large amount of time manually searching multiple portals for internships that fit their skill set, and often prepare for interviews without personalized, role-specific guidance. This project addresses that gap by using AI to match candidates to relevant internships and to generate tailored interview preparation material based on the candidate's profile and the target role.

1.4 Target Users

- ☐ Students
- ☐ Freshers / early-career job seekers
- ☐ Career switchers
- ☐ Campus placement cells / mentors

2. Project Objective

The project aims to achieve the following objectives:

- ☐ Automate internship discovery by matching a candidate's skills and profile to relevant openings.
- ☐ Provide an AI-based interview preparation module with role-specific questions and feedback.
- ☐ Reduce the time and effort required to find suitable internship opportunities.
- ☐ Deliver personalized, data-driven career guidance rather than generic advice.

3. Technology Stack

Category	Technology
Programming Language	Python
Backend	FastAPI
Frontend	React
AI / LLM	OpenAI / Gemini / Groq
AI Framework	LangChain / LangGraph

Category	Technology
Database	PostgreSQL
Vector Database	ChromaDB / FAISS
Deployment	Docker / Cloud
Version Control	Git / GitHub

Note: This table lists a representative stack for an AI career-matching project. Please confirm against the actual repository (<https://github.com/MallarpuAishwarya/AI-Career-Companion-Agent-for-Internship-Matching-and-Interview-Preparation>) and remove/replace any technology not actually used.

4. Project Requirements

Functional Requirements

- ☐ User should be able to register/login.
- ☐ User should be able to upload a resume / enter skill profile.
- ☐ System should match the user's profile to relevant internship listings.
- ☐ User should be able to request interview preparation for a target role.
- ☐ System should generate role-specific interview questions and sample answers.
- ☐ System should store conversation and preparation history.

Non-Functional Requirements

- ☐ Good response time
- ☐ Security
- ☐ Scalability
- ☐ User-friendly interface
- ☐ Error handling

5. System Architecture / Workflow

This section explains how the complete system works end to end.

- ☐ Architecture diagram
- ☐ Main components
- ☐ Data flow
- ☐ API flow
- ☐ AI / RAG / Agent workflow

High-Level Flow

User → React Frontend → FastAPI Backend → Matching / RAG Pipeline → LLM → Response → User

AI / Matching Workflow

Resume / Skills → Text Extraction → Chunking & Embeddings → Vector DB → Retrieval → LLM Reasoning → Internship Match / Interview Questions

6. Product Backlog

ID	Feature / Task	Priority	Status
P001	Project Setup	High	Done
P002	Database Setup	High	Done
P003	User Authentication	High	Done
P004	Resume / Profile Upload	High	Done
P005	Internship Matching Engine	High	In Progress
P006	AI Interview Question Generator	High	In Progress
P007	Chat Interface	Medium	To Do
P008	Testing	High	To Do

7. Sprint Planning

Sprint 1

Duration: 2 Weeks (15 Jul – 28 Jul)

Goal: Project Setup

- ☐ Understand requirements
- ☐ Create project structure
- ☐ Set up GitHub repository
- ☐ Set up database
- ☐ Set up development environment

Sprint 2

Duration: 2 Weeks (29 Jul – 11 Aug)

Goal: Backend Development

- ☐ Create APIs
- ☐ Implement authentication
- ☐ Create database models
- ☐ Implement resume / profile upload

Sprint 3

Duration: 2 Weeks (12 Aug – 25 Aug)

Goal: AI / Matching Implementation

- ☐ Resume / JD processing
- ☐ Text chunking
- ☐ Embedding generation
- ☐ Vector database setup
- ☐ LLM integration
- ☐ Matching & interview-question pipeline

Sprint 4

Duration: 2 Weeks (26 Aug – 9 Sep)

Goal: Final Integration & Testing

- ☐ Frontend integration
- ☐ API testing
- ☐ Feature testing
- ☐ Bug fixing
- ☐ Final demo preparation

8. Sprint Tasks & Status

Sprint	Task	Assigned To	Status
Sprint 1	Project Setup	Mallarpu Aishwarya	Done
Sprint 1	Database Setup	Mallarpu Aishwarya	Done
Sprint 2	API Development	Mallarpu Aishwarya	In Progress
Sprint 2	UI Development	Mallarpu Aishwarya	Testing
Sprint 3	Matching / RAG Implementation	Mallarpu Aishwarya	In Progress

Task Status Flow: To Do → In Progress → Testing → Done

9. Implementation / Key Features

Feature 1: User Authentication

Description: Allows a user to securely register and log in to the platform.

Implementation: Implemented using the backend authentication flow described in the repository.

Output: [Add screenshot / sample response]

Feature 2: Resume / Profile Upload & Internship Matching

Description: User uploads a resume or enters skills; the system matches the profile against internship listings.

Implementation: Explain the extraction, embedding, and matching flow used in the project.

Output: [Add screenshot / sample response]

Feature 3: AI Interview Preparation

Description: User selects a target role and receives AI-generated, role-specific interview questions and guidance.

Implementation: Explain the LLM/RAG workflow used to generate and personalize the questions.

Output: [Add screenshot / sample response]

Note: Do not paste large amounts of source code here. Show: Feature → How it works → Screenshot → Result.

10. Testing

Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
T001	Login with valid credentials	User logs in	User logged in	Pass
T002	Upload resume	Resume uploaded	Resume uploaded	Pass
T003	Request internship match	Relevant matches returned	Matches returned	Pass
T004	Request interview questions	Role-specific questions returned	Questions returned	Pass
T005	Upload invalid file	Error message shown	Error shown	Pass

Testing Types

- ☐ Functional Testing
- ☐ API Testing
- ☐ UI Testing
- ☐ Integration Testing
- ☐ AI / Matching Testing

11. Challenges & Solutions

Challenge	Solution
Resume text extraction issues	Used an appropriate document-processing library to reliably parse varied resume formats.

Challenge	Solution
Slow API response	Optimized API and database operations, and cached repeated LLM calls where possible.
Irrelevant or inaccurate matches	Improved retrieval strategy and refined prompts used for matching and question generation.
Integration issues	Tested frontend, backend, and AI components independently before full integration.

Note: Replace the entries above with the real challenges you personally faced during development — this section should reflect your actual experience, not generic problems.

12. Final Project Status

Completed Features

- ✓ [Feature 1]
- ✓ [Feature 2]
- ✓ [Feature 3]
- ✓ [Feature 4]

In Progress

- ☐ [Feature]

Pending

- ☐ [Feature]

Final Outcome

The project implements resume/profile-based internship matching and AI-driven interview preparation. Core features have been developed and tested; remaining work includes UI polish, further testing, and deployment. [Update this summary with the project's actual current status.]

Project Repository

<https://github.com/MallarpuAishwarya/AI-Career-Companion-Agent-for-Internship-Matching-and-Interview-Preparation>